GPSC CBRT · HOMOEOPATHIC PHYSICIAN · CORE

Topic 7: Practice of Medicine & Homoeopathic Therapeutics

Complete notes with pictures, memory hooks, rapid revision and practice MCQs.

Topic 7: Practice of Medicine & Homoeopathic Therapeutics

How to use this in 2 days

Topic 7 is the largest clinical topic in the core: expect about 3 of the 25 core marks, plus medicine-flavoured questions hidden under pathology and community medicine. GPSC asks the diagnostic number, the drug of choice, the classic sign and one keynote remedy per disease.

Session	Read	Then do
Day 1 morning	GPSC questions already asked; hypertension and IHD; asthma and COPD	Write the five cut-off numbers from memory
Day 1 afternoon	Stroke and epilepsy; GI (peptic ulcer, GERD, IBS); liver and gallbladder	First 25 practice MCQs
Day 1 evening	Diabetes; thyroid, gout, nutrition	Rapid revision 1–30
Day 2 morning	Skin (eczema, psoriasis, urticaria); fevers and seasonal outbreaks	—
Day 2 afternoon	Homoeopathic therapeutics table, twice	Last 25 practice MCQs
Day 2 evening	Rapid revision page; your wrong answers	Sleep

How GPSC frames medicine:

- **Numbers**: BP 140/90, fasting glucose 126, HbA1c 6.5%, FEV1/FVC 0.7, thrombolysis 4.5 h.
- **'Investigation of choice' and 'drug of choice'**: CT for acute stroke/SAH, spirometry for COPD, PTU in first-trimester hyperthyroidism, salbutamol for acute asthma.
- **'All EXCEPT' and 'which is FALSE'**: the 2025 Medical Officer paper opened with one on diabetes.
- **Homoeopathic therapeutics**: one disease, one leading remedy, one keynote. The therapeutics table near the end is built for this.

What GPSC has already asked

The Medical Officer papers of 28 Sep 2024 and 28 Sep 2025 asked 17 medicine questions between them. All are one-fact questions: a target, a drug of choice, a mechanism, a pattern. The answers are GPSC's keys, checked against standard texts.

Paper	Question (shortened)	GPSC answer	The fact to carry away
MO 2025	Diabetes: which statement is false?	Type 2 more prone to ketoacidosis than type 1	Type 1 is ketosis-prone; type 2 deaths are mostly cardiovascular; FBS ≥ 126 diagnoses
MO 2025	Vitamin most likely deficient in strict vegans	Vitamin B12	B12 comes only from animal foods
MO 2025	Antithyroid drug safer in the 1st trimester	Propylthiouracil	Switch to carbimazole/methimazole after the 1st trimester (PTU liver toxicity)
MO 2025	Non-modifiable risk factor for hypertension	Gender	Also age, family history, ethnicity
MO 2025	Investigation of choice in acute SAH	Non-contrast CT	LP for xanthochromia if CT negative
MO 2025	RHD with AF, sudden left hemiparesis, infarct on CT	Embolic stroke	AF is the commonest source of cardio-embolism
MO 2025	Loss of smell and taste with cough (2020)	Nasopharyngeal swab RT-PCR	Reference test for SARS-CoV-2
MO 2025	Contraindication to parenteral nutrition	Active infection	GPSC's key; uncontrolled sepsis is a relative contraindication
MO 2025	Kwashiorkor differs from marasmus by	Oedema	Protein deficiency with oedema; marasmus = severe wasting
MO 2024	BP goal in a diabetic with hypertension	120–130 / 70–79 mmHg	Aim < 130/80 in diabetes
MO 2024	True about furosemide	Indicated in acute pulmonary oedema	Loop diuretic (thick ascending limb); potent; oral and IV
MO 2024	Key mechanism in GERD	Transient LES relaxations	Not hiatus hernia, not LES hypertension
MO 2024	Commonest pattern in IBS	Constipation alternating with diarrhoea	Rome IV: pain related to defecation
MO 2024	Graves' disease thyroid profile	Low TSH	High T3/T4, TSH-receptor antibodies
MO 2024	Statin effective at any time of day	Atorvastatin	Long half-life (rosuvastatin too); simvastatin and lovastatin at night

Paper	Question (shortened)	GPSC answer	The fact to carry away
MO 2024	Leukonychia	Hypoalbuminaemia	Koilonychia = iron deficiency; clubbing = lung/heart disease
MO 2024	Mood stabiliser with anti-suicidal action	Lithium	Narrow therapeutic range; check levels, thyroid and renal function

Pattern: GPSC writes long clinical stems but asks one textbook fact. Read the last line of the stem first.

Hypertension and ischaemic heart disease

Hypertension is diagnosed at **≥ 140/90 mmHg** on repeated clinic readings (Indian and ISH/WHO guidelines); ACC/AHA 2017 uses ≥ 130/80. Troponin is the marker of choice for myocardial infarction.

Hypertension

Point	Fact
Grades (ISH 2020)	Grade 1: 140–159 / 90–99; Grade 2: ≥ 160 / ≥ 100
Commonest type	Essential (primary), ~90–95%
Secondary causes	Renal (commonest secondary: renal parenchymal disease), renal artery stenosis, Cushing's, Conn's (low K+), phaeochromocytoma (episodic headache, sweating, palpitations), coarctation (radio-femoral delay), OCPs, NSAIDs
Non-modifiable risks	Age, gender, family history, ethnicity (MO 2025)
Modifiable risks	Salt > 5 g/day, obesity, alcohol, inactivity, stress, smoking
Targets	General < 140/90 (< 130/80 if tolerated); diabetes or CKD < 130/80 (MO 2024)
Lifestyle	DASH diet, salt < 5 g/day (WHO), weight loss, exercise 150 min/week
First-line drugs	ACE inhibitor/ARB, calcium-channel blocker (amlodipine), thiazide-like diuretic
In pregnancy	Labetalol, nifedipine, methyldopa; **ACE inhibitors/ARBs are contraindicated**
Hypertensive emergency	BP ≥ 180/120 with organ damage: IV labetalol, nitroprusside, nicardipine; lower by no more than 25% in the first hour
Target-organ damage	LVH, retinopathy (Keith–Wagener grades; papilloedema = grade IV), nephrosclerosis, stroke

Drug side-effects asked often: ACE inhibitors → **dry cough** (bradykinin), angio-oedema, hyperkalaemia; amlodipine → ankle oedema; thiazides → hypokalaemia, hyperuricaemia, hyperglycaemia; beta-blockers → avoid in asthma.

Ischaemic heart disease

Condition	Key features	Diagnosis / treatment
Stable angina	Exertional retrosternal pain < 10 min, relieved by rest or sublingual nitrate	ECG stress test; coronary angiography = gold standard
Unstable angina / NSTEMI	Pain at rest or crescendo; NSTEMI has raised troponin	Antiplatelets, anticoagulant, early angiography
STEMI	Pain > 20 min, sweating; ST elevation in contiguous leads or new LBBB	**Primary PCI** within 90–120 min, or thrombolysis (streptokinase, tenecteplase) within 12 h
Prinzmetal (variant) angina	Rest pain from coronary spasm, transient ST elevation	Calcium-channel blockers, nitrates

- **Biomarkers**: troponin I/T most sensitive and specific (rises 3–4 h, stays 7–10 days); CK-MB rises 4–6 h, normal by 48–72 h (useful for re-infarction).
- **Infarct territory on ECG**: II, III, aVF = inferior (right coronary); V1–V4 = anterior (LAD); I, aVL, V5–V6 = lateral (circumflex).
- **MONA-B immediate care**: aspirin 300 mg chewed (all patients), clopidogrel/ticagrelor, nitrates (avoid in right ventricular infarct or hypotension), oxygen only if saturation < 90%, morphine for pain; plus statin and beta-blocker.
- **Complications**: arrhythmia (VF — commonest cause of early death), heart failure, cardiogenic shock, papillary-muscle rupture (mitral regurgitation), septal rupture, free-wall rupture (days 3–7), Dressler syndrome (weeks later, pericarditis).
- **Secondary prevention**: aspirin, statin (high-intensity: atorvastatin 40–80 mg; any time of day — MO 2024), beta-blocker, ACE inhibitor, stop smoking.
- **Heart failure basics**: breathlessness, orthopnoea, PND, raised JVP, basal crepitations, pedal oedema; BNP/NT-proBNP raised; echo for ejection fraction; furosemide for congestion and acute pulmonary oedema (MO 2024).

Memory hook: **A**CE inhibitors give an **A**nnoying cough (bradykinin). After an MI, **troponin is the tortoise**: it rises in a few hours and stays for about a week or two; **CK-MB is the hare**: gone in 2 to 3 days, so it is the marker for a fresh re-infarction.

Bronchial asthma and COPD

The one-line difference: asthma is **variable and reversible** airflow obstruction (inflammation, often allergic, young); COPD is **persistent, not fully reversible** obstruction (smoking or biomass smoke, older).

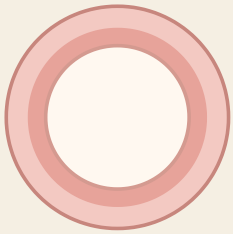
Feature	Bronchial asthma	COPD
Onset	Often childhood, atopic	> 40 years, smoker
Symptoms	Episodic wheeze, chest tightness, night/early-morning cough	Chronic productive cough, progressive breathlessness
Triggers	Allergens (house dust mite, pollen), cold air, exercise, aspirin/NSAIDs, beta-blockers, infections	Smoking (commonest cause), biomass fuel (common in Indian women), occupational dust
Spirometry	Reversibility: FEV1 rises ≥ 12% and ≥ 200 mL after bronchodilator; PEF variability > 10%	**Post-bronchodilator FEV1/FVC < 0.70** (GOLD)
Cells	Eosinophils, mast cells, Th2	Neutrophils, macrophages, CD8
Sputum/pathology	Curschmann spirals, Charcot–Leyden crystals, Creola bodies	Goblet-cell hyperplasia; Reid index > 0.4 (chronic bronchitis)
Imaging	Usually normal	Hyperinflation, flat diaphragm, bullae (emphysema)

Definitions

- **Chronic bronchitis**: productive cough for ≥ 3 months a year for 2 consecutive years ('blue bloater').
- **Emphysema**: permanent enlargement of airspaces distal to terminal bronchioles with wall destruction ('pink puffer'). Centriacinar = smokers (upper lobes); panacinar = **alpha-1 antitrypsin deficiency** (lower lobes, young).
- **Status asthmaticus / acute severe asthma**: PEF 33–50% predicted, can't complete sentences, pulse ≥ 110, RR ≥ 25. **Life-threatening**: PEF < 33%, SpO2 < 92%, silent chest, cyanosis, normal or rising PaCO2.

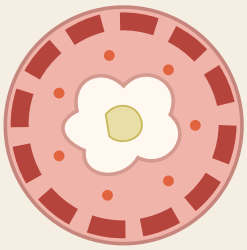
Asthma and COPD in the airway

What narrows the airway, and what destroys the alveoli



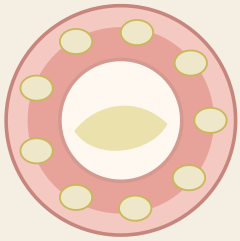
Normal airway

Open lumen, thin wall



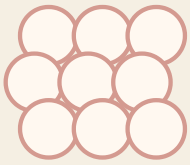
Asthma

Smooth muscle squeezes, wall swells, mucus plug; eosinophils, mast cells



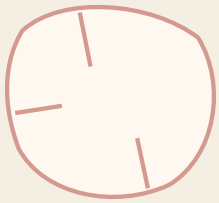
Chronic bronchitis

Enlarged mucus glands, goblet cells (Reid index > 0.4); "blue bloater"



Normal alveoli

Many small sacs, large surface for gas exchange



Emphysema

Walls destroyed, airspaces enlarged; "pink puffer"

Centriacinar: smokers, upper lobes
Panacinar: α-1 antitrypsin, lower lobes

Asthma: variable and REVERSIBLE (FEV1 rises ≥ 12% and ≥ 200 mL after a bronchodilator).

COPD: persistent, not fully reversible (post-bronchodilator FEV1/FVC < 0.70); smoking is the commonest cause.

Asthma and COPD · what narrows the airway, and what destroys the alveoli

Treatment

Situation	Treatment
Acute asthma attack	Inhaled **short-acting beta-2 agonist (salbutamol)** by nebuliser/spacer + ipratropium; oxygen to SpO2 93–95%; systemic steroid (prednisolone/hydrocortisone); IV magnesium sulphate if severe
Long-term asthma (GINA)	Inhaled corticosteroid is the controller; ICS–formoterol as reliever and controller; add LABA, then LAMA or montelukast; anti-IgE (omalizumab) in severe allergic asthma
COPD maintenance	Long-acting bronchodilators: LAMA (tiotropium) and/or LABA; ICS only if frequent exacerbations with eosinophilia

Situation	Treatment
COPD exacerbation	Bronchodilators, steroids, antibiotics if purulent sputum; **controlled oxygen (target SpO2 88–92%)** to avoid CO2 retention; NIV for hypercapnic acidosis
Proven to improve COPD survival	**Smoking cessation** and **long-term oxygen therapy** (≥ 15 h/day if PaO2 ≤ 55 mmHg)
Vaccines	Influenza yearly; pneumococcal

Complications of COPD: cor pulmonale (right heart failure, raised JVP, oedema), secondary polycythaemia, type 2 respiratory failure (raised CO2: bounding pulse, flapping tremor, headache), pneumothorax from bullae.

- Related respiratory facts often clubbed in: **pulmonary TB** — cough > 2 weeks, fever, night sweats, weight loss, haemoptysis; NAAT (CBNAAT/TrueNat) is the initial test under NTEP.
- **Community-acquired pneumonia** — commonest organism Streptococcus pneumoniae; CURB-65 for severity.
- **Bronchiectasis** — copious foul sputum; HRCT is investigation of choice; 'signet ring' sign.

Memory hook: **A**sthma comes and goes **A**gain (variable, reversible); **COPD Clings** (persistent). An FEV1/FVC ratio **below 0.7** means obstruction; a rise after a bronchodilator means asthma.

Stroke and epilepsy

Non-contrast CT is the first investigation in every acute stroke: it separates haemorrhage from infarction before any clot-buster is given.

Stroke

Point	Fact
Definition	Sudden focal neurological deficit of vascular origin lasting > 24 h or causing death; TIA resolves without infarction (most within 1 h)
Types	Ischaemic ~85% (thrombotic, embolic, lacunar); haemorrhagic ~15% (intracerebral, subarachnoid)
Commonest risk factor	**Hypertension**; then diabetes, smoking, AF, dyslipidaemia
Commonest cardiac source of emboli	**Atrial fibrillation** (rheumatic mitral stenosis in India) — MO 2025 case

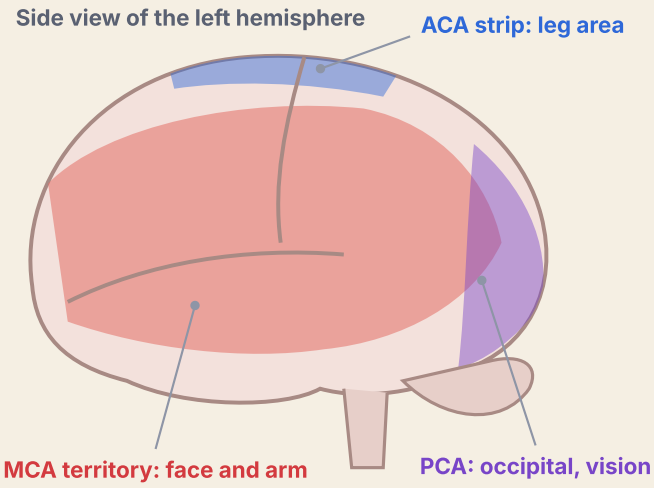
Point	Fact
Commonest site	Middle cerebral artery: contralateral hemiplegia (face and arm > leg), aphasia if dominant side
ACA stroke	Leg > arm weakness
Internal capsule	Dense hemiplegia of face, arm and leg (lenticulostriate arteries; hypertensive haemorrhage)
Brainstem / PCA	Crossed signs; vertigo, diplopia; homonymous hemianopia with macular sparing (PCA)
Investigation of choice	**Non-contrast CT** (haemorrhage white/hyperdense); MRI-DWI best for early infarct
Thrombolysis	IV alteplase/tenecteplase within 4.5 h of onset if CT excludes bleed; mechanical thrombectomy up to 6–24 h for large-vessel occlusion
After the acute phase	Aspirin, statin, BP control; anticoagulate for AF (warfarin/DOAC)
Scores	FAST for recognition (Face, Arm, Speech, Time); NIHSS for severity; GCS for consciousness (3–15)

Subarachnoid haemorrhage: 'worst headache of my life' (thunderclap), neck stiffness, photophobia; commonest cause **ruptured berry aneurysm** (circle of Willis, anterior communicating artery). CT first (MO 2025); if negative, lumbar puncture after 12 h for **xanthochromia**. Nimodipine prevents vasospasm.

Bell's palsy (LMN facial palsy): whole half of face including forehead; steroids within 72 h. In UMN (stroke) the forehead is spared.

Stroke territories and facial palsy

Which artery, which weakness; does the forehead move?



Middle cerebral artery
COMMONEST site. Opposite-side weakness, face and arm > leg; aphasia if dominant side

Anterior cerebral artery
Leg > arm weakness

Posterior cerebral artery
Homonymous hemianopia, macular sparing

Internal capsule
Dense weakness of face, arm AND leg (lenticulostriate; hypertensive bleed)

Stroke (UMN facial palsy)



Lower face droops on one side
FOREHEAD SPARED: wrinkles on both sides

Bell's palsy (LMN)



WHOLE half of the face, forehead too: no wrinkles, eye will not close.
Steroids within 72 h

Non-contrast CT first in every acute stroke: it rules out bleeding before a clot-buster.

Stroke territories and facial palsy · MCA face and arm, ACA leg; is the forehead spared?

Epilepsy

Seizure type	Features	Drug of choice
Generalised tonic-clonic	Tonic then clonic phase, tongue bite, incontinence, post-ictal confusion	Valproate (avoid in women of child-bearing age → levetiracetam/lamotrigine)
Absence (petit mal)	Children; brief staring spells; **3 Hz spike-and-wave** EEG; hyperventilation provokes	**Ethosuximide** or valproate

Seizure type	Features	Drug of choice
Focal (partial)	Aura, automatisms (temporal lobe: déjà vu, lip-smacking)	Carbamazepine or lamotrigine
Myoclonic (juvenile myoclonic epilepsy)	Morning jerks, teens	Valproate (carbamazepine worsens it)
Infantile spasms (West syndrome)	Flexor spasms, hypsarrhythmia on EEG	ACTH, vigabatrin
Febrile seizure	6 months–5 years, with fever, usually simple and benign	Treat fever; no long-term drugs

Status epilepticus: seizure > 5 minutes or repeated seizures without recovery. Treatment: airway, glucose, then **IV lorazepam** (or midazolam/diazepam), then phenytoin/fosphenytoin, valproate or levetiracetam; if refractory, anaesthesia.

In India the **commonest cause of acquired seizures is neurocysticercosis** (Taenia solium; ring-enhancing lesion on CT). Investigations: EEG, MRI brain. Pregnancy: valproate is the most teratogenic (neural-tube defects); give folic acid. Phenytoin side-effects: gum hypertrophy, hirsutism, ataxia, nystagmus.

Memory hook: plain CT first: it rules out bleeding before any clot-busting drug. "Worst headache of my life" = subarachnoid haemorrhage. In facial palsy, an **upper** motor neurone lesion **spares the forehead**; **Bell's palsy** (lower motor neurone) takes the whole half-face.

Peptic ulcer, GERD and irritable bowel syndrome

H. pylori and NSAIDs cause most peptic ulcers; transient LES relaxation causes GERD; IBS is a functional disorder diagnosed by the Rome IV criteria with no structural disease.

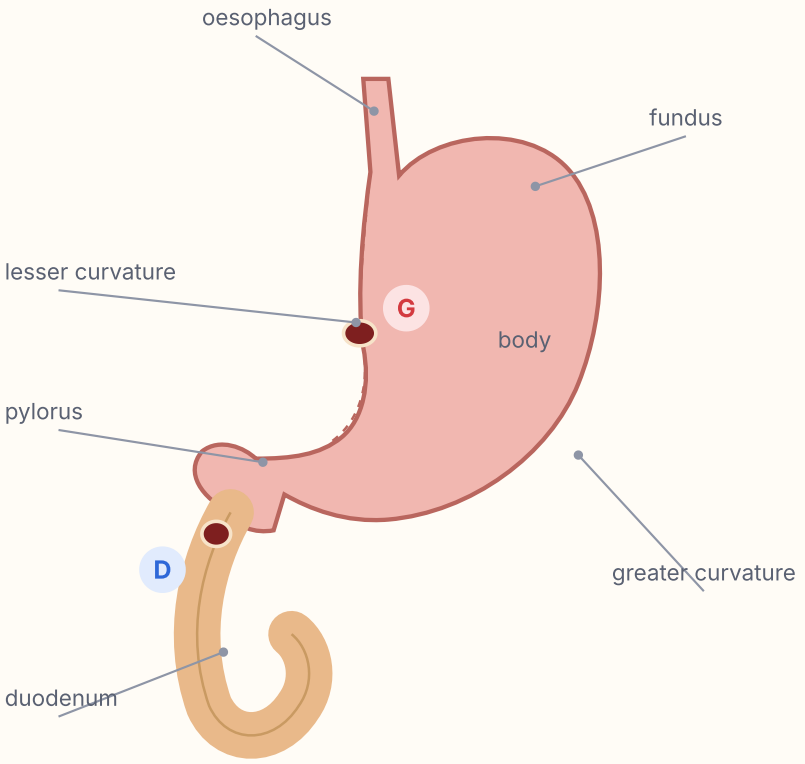
Peptic ulcer disease

Point	Duodenal ulcer	Gastric ulcer
Frequency	Commoner (about 4:1)	Less common
Pain	Epigastric, **relieved by food**, hunger pain, wakes at night	**Worse after food**, weight loss

Point	Duodenal ulcer	Gastric ulcer
Cause	H. pylori in ~90%; high acid	H. pylori ~70%, NSAIDs
Cancer risk	Almost never malignant	Biopsy all gastric ulcers (can be malignant)
Site	First part of duodenum, anterior wall commonest	Lesser curvature

Peptic ulcer: duodenal or gastric?

Site, cause and how food changes the pain



D Duodenal ulcer

Commoner (about 4 : 1)
1st part of duodenum,
anterior wall commonest
H. pylori in ~90%; high acid
Almost never malignant

G Gastric ulcer

Lesser curvature
H. pylori ~70%, NSAIDs
Biopsy every gastric ulcer:
it can be malignant

PAIN AND FOOD

Duodenal: relieved by food, hunger pain, wakes at night

Gastric: worse after food, weight loss

"Gastric Gets worse with food;
Duodenal is Dulled by food."

Peptic ulcer · duodenal vs gastric: site, cause and the effect of food

- H. pylori tests: **urea breath test** and stool antigen (non-invasive; best for confirming eradication), rapid urease (CLO) test on biopsy, histology.
- Eradication: PPI + clarithromycin + amoxicillin (or metronidazole) for 14 days; bismuth quadruple therapy if resistance.
- Complications: **bleeding (commonest)**, perforation (anterior duodenal ulcer; air under diaphragm), gastric outlet obstruction (projectile non-bilious vomiting, succussion splash, hypokalaemic hypochloraemic metabolic alkalosis), penetration (posterior ulcer into pancreas).

- Zollinger–Ellison syndrome: gastrinoma, multiple/refractory ulcers, diarrhoea; raised fasting gastrin.

GERD

- Mechanism: **transient lower oesophageal sphincter relaxations** (MO 2024); hiatus hernia, obesity, pregnancy, smoking, alcohol, fatty food contribute.
- Symptoms: heartburn, acid regurgitation worse after meals and lying down; atypical: chronic cough, hoarseness, asthma, dental erosion.
- Diagnosis: clinical trial of PPI; endoscopy for alarm features (dysphagia, weight loss, bleeding, anaemia, age > 60); **24-hour pH monitoring = gold standard**.
- Treatment: lifestyle (raise head end, avoid late meals, lose weight), **PPIs** (most effective), H2 blockers; surgery = Nissen fundoplication.
- Complications: oesophagitis, stricture, **Barrett's oesophagus** (intestinal metaplasia) → adenocarcinoma.
- Oesophagus contrasts: achalasia = **bird's beak**, dysphagia to solids and liquids from the start; carcinoma = **rat tail** (MO 2025), progressive dysphagia solids → liquids.

Irritable bowel syndrome

- **Rome IV**: recurrent abdominal pain on average ≥ 1 day/week for the last 3 months, with ≥ 2 of: related to defecation, change in stool frequency, change in stool form; onset ≥ 6 months before diagnosis.
- Types: IBS-C, IBS-D, IBS-M (mixed); the commonest pattern keyed by GPSC is **constipation alternating with diarrhoea** (MO 2024).
- Features that are NOT IBS (red flags): age > 50 at onset, rectal bleeding, weight loss, nocturnal diarrhoea, anaemia, family history of colon cancer.
- Management: reassurance, diet (low-FODMAP), fibre for constipation, antispasmodics (mebeverine, dicyclomine), loperamide for diarrhoea, low-dose tricyclics; stress is a major trigger.

Inflammatory bowel disease (for contrast)

Ulcerative colitis	Crohn's disease
Rectum upwards, continuous	Mouth to anus, skip lesions; terminal ileum commonest
Mucosal; bloody diarrhoea	Transmural; fistulae, strictures, non-caseating granulomas
Lead-pipe colon; pseudopolyps	String sign of Kantor; cobblestone mucosa

Ulcerative colitis	Crohn's disease
Smoking protects; p-ANCA	Smoking worsens; ASCA
Higher colon-cancer risk; toxic megacolon	Perianal disease, malabsorption (B12)

Memory hook: **G**astric ulcer pain **G**ets worse with food; **D**uodenal ulcer pain is **D**ulled by food and wakes the patient at night. Most ulcers trace back to **H. pylori** or **NSAIDs**.

Hepatitis, cirrhosis and cholecystitis

Viral hepatitis transmission and serology, the complications of cirrhosis, and Murphy's sign with ultrasound carry most marks here.

Viral hepatitis

Virus	Route	Chronicity	Clinical point
A	Faeco-oral	None	Children, mild; anti-HAV IgM = acute; vaccine
B (DNA)	Blood, sexual, perinatal	5–10% adults; ~90% of infected newborns	HBsAg = infection; anti-HBc IgM = acute; anti-HBs = immunity; vaccine at birth (UIP)
C	Blood (transfusion, IV drugs, dialysis)	~80%	Cirrhosis, HCC; cured by direct-acting antivirals (sofosbuvir); no vaccine
D	Needs HBV	—	Superinfection worsens HBV
E	Faeco-oral, water-borne epidemics	Rare	**~20% mortality in pregnancy** (fulminant failure)

Acute hepatitis: prodrome (anorexia, nausea, distaste for smoking), then jaundice; **ALT > AST**, often > 1000 U/L. Alcoholic hepatitis: **AST:ALT > 2**, raised GGT. Fulminant hepatic failure: encephalopathy within 8 weeks, prolonged INR (best prognostic marker). Paracetamol overdose antidote: **N-acetylcysteine**.

Cirrhosis

- Causes: alcohol, chronic HBV/HCV, **NAFLD/MASLD** (rising), autoimmune hepatitis, haemochromatosis, Wilson's disease (Kayser–Fleischer rings, low caeruloplasmin), primary biliary cholangitis (anti-mitochondrial antibody).

- Signs: palmar erythema, spider naevi (SVC area), gynaecomastia, testicular atrophy, Dupuytren's contracture (alcohol), clubbing, leukonychia (low albumin; MO 2024), jaundice, ascites, caput medusae, splenomegaly.
- Severity: **Child–Pugh** (bilirubin, albumin, INR, ascites, encephalopathy); MELD for transplant listing.

Complication	Key facts
Portal hypertension, oesophageal varices	Haematemesis; endoscopic band ligation, IV octreotide/terlipressin; non-selective beta-blocker (propranolol) prevents bleeds
Ascites	SAAG ≥ 1.1 g/dL = portal hypertension; salt restriction, **spironolactone** + furosemide; large-volume paracentesis with albumin
Spontaneous bacterial peritonitis	Ascitic neutrophils > 250/mm^3; cefotaxime
Hepatic encephalopathy	**Asterixis (flapping tremor)**, confusion, fetor hepaticus; precipitated by GI bleed, infection, constipation, sedatives, high protein; **lactulose**, rifaximin
Hepatorenal syndrome	Functional renal failure; terlipressin + albumin
Hepatocellular carcinoma	Raised **AFP**; ultrasound + AFP surveillance every 6 months

Gallbladder: cholelithiasis and cholecystitis

- Gallstone risk: '**fat, female, fertile, forty, fair**'; cholesterol stones commonest; pigment stones in haemolysis.
- **Biliary colic**: right upper quadrant/epigastric pain after fatty meals, radiating to right scapula.
- **Acute cholecystitis**: persistent pain, fever, **Murphy's sign** (inspiration arrested on palpating the right subcostal area); leucocytosis.
- **Investigation of choice: ultrasound** (stones with acoustic shadow, wall thickening, pericholecystic fluid). HIDA scan if doubt.
- Treatment: analgesia, antibiotics, **laparoscopic cholecystectomy** (early, within 72 h ideally).
- Complications: empyema, gangrene, perforation, **Mirizzi syndrome**, gallstone ileus.
- **Charcot's triad** (ascending cholangitis): fever with rigors, jaundice, RUQ pain; add hypotension and confusion = **Reynolds' pentad**.
- **Courvoisier's law**: a palpable, non-tender gallbladder with jaundice is unlikely to be due to stones — think carcinoma head of pancreas.
- Acute pancreatitis: epigastric pain radiating to the back, relieved by leaning forward; **lipase** (more specific) / amylase > 3× normal; commonest causes gallstones and alcohol; Cullen's (periumbilical) and Grey Turner's (flank) signs.

Memory hook: viral hepatitis lifts **ALT above AST**; alcohol flips it, **AST:ALT above 2** ("**S**cotch makes A**S**T soar"). Gallstones love the **5 Fs**: fat, female, fertile, forty, fair.

Diabetes mellitus

Diagnosis rests on four cut-offs: fasting plasma glucose ≥ 126 mg/dL, 2-h OGTT ≥ 200, HbA1c ≥ 6.5%, or random ≥ 200 with symptoms. Type 1 is ketosis-prone; type 2 patients die mostly of cardiovascular disease.

Test	Normal	Prediabetes	Diabetes
Fasting plasma glucose (mg/dL)	< 100	100–125 (IFG)	≥ 126
2-h OGTT, 75 g (mg/dL)	< 140	140–199 (IGT)	≥ 200
HbA1c	< 5.7%	5.7–6.4%	≥ 6.5%
Random glucose	—	—	≥ 200 with classic symptoms

Diabetes starts at 126, 200 and 6.5

Plasma glucose in mg/dL and HbA1c in %; read each row from green to red

	Normal	Prediabetes	Diabetes
Fasting glucose	below 100	100 to 125	126 or more
2-hour glucose (OGTT)	below 140	140 to 199	200 or more
HbA1c	below 5.7	5.7 to 6.4	6.5 or more

A random glucose of 200 or more with classic symptoms is also diagnostic.

diabetes cut-offs · normal, prediabetes, diabetes

The red column is the one examiners ask: 126, 200, 6.5.

An abnormal result must be repeated on another day unless symptoms are clear-cut.

Type 1 vs type 2

Type 1	Type 2
Autoimmune β-cell destruction (GAD, ICA antibodies); HLA-DR3/DR4	Insulin resistance + relative deficiency; obesity, family history (stronger genetic link)

Type 1	Type 2
Young, thin, abrupt onset	Usually > 30–40 years, obese, insidious (Indians at younger age and lower BMI)
Prone to diabetic ketoacidosis	Prone to hyperosmolar hyperglycaemic state
Low C-peptide; insulin essential	Lifestyle + metformin first line

Other types: gestational diabetes (India: DIPSI single-step 75 g OGTT, 2-h ≥ 140 mg/dL); secondary diabetes from **acromegaly, Cushing's syndrome**, phaeochromocytoma, glucagonoma, chronic pancreatitis, steroids (all endocrine causes asked in MO 2025); MODY.

Acute complications

Condition	Features	Treatment
Diabetic ketoacidosis	Glucose > 250, pH < 7.3, HCO3 < 18, ketones; Kussmaul breathing, fruity breath, abdominal pain, dehydration	**IV normal saline first**, then insulin infusion 0.1 U/kg/h; potassium replacement (check K before insulin)
Hyperosmolar hyperglycaemic state	Older type 2; glucose > 600, osmolality > 320, little ketosis, severe dehydration, confusion	Fluids, cautious insulin
Hypoglycaemia	Glucose < 70 mg/dL; sweating, tremor, confusion; Whipple's triad	15–20 g oral glucose; IV 25% dextrose or glucagon if unconscious

Chronic complications: microvascular — retinopathy (commonest cause of blindness in working adults; screen yearly), nephropathy (earliest sign **microalbuminuria 30–300 mg/day**; Kimmelstiel–Wilson nodules), neuropathy (glove-and-stocking, diabetic foot); macrovascular — coronary disease (commonest cause of death), stroke, peripheral vascular disease.

Treatment and targets

- Targets: HbA1c < 7% for most adults; BP < 130/80 (MO 2024); LDL < 100 (< 70 with heart disease); statin for most diabetics over 40.
- **Metformin** = first-line oral drug; does not cause hypoglycaemia; side-effects GI upset, B12 deficiency, **lactic acidosis** (avoid if eGFR < 30).
- Sulphonylureas (glimepiride, gliclazide): hypoglycaemia, weight gain. DPP-4 inhibitors (sitagliptin): weight-neutral. **SGLT2 inhibitors** (empagliflozin, dapagliflozin): heart-failure and kidney protection;

genital infections, euglycaemic DKA. GLP-1 agonists (liraglutide, semaglutide): weight loss. Pioglitazone: fluid retention, avoid in heart failure.

- Insulins: rapid (lispro, aspart), short (regular), intermediate (NPH), long-acting (glargine, degludec). Insulin is the drug of choice in pregnancy.
- Somogyi effect = morning hyperglycaemia after night hypoglycaemia; dawn phenomenon = early-morning rise from growth hormone and cortisol.

Memory hook: diabetes cut-offs read **126, 200, 6.5**: fasting glucose 126 mg/dL or more, 2-hour (or random with symptoms) 200 mg/dL or more, HbA1c 6.5% or more.

Thyroid disorders, gout and nutritional deficiencies

TSH is the best single screening test for thyroid function; serum uric acid alone does not diagnose gout (crystals do); and India still carries both deficiency diseases and obesity.

Thyroid

Condition	Labs	Key features	Treatment
Primary hypothyroidism	**High TSH, low free T4**	Weight gain, cold intolerance, constipation, dry skin, bradycardia, menorrhagia, delayed relaxation of ankle jerk; commonest cause worldwide = iodine deficiency, in iodine-sufficient areas = **Hashimoto's** (anti-TPO)	Levothyroxine on an empty stomach; recheck TSH after 6–8 weeks
Subclinical hypothyroidism	High TSH, normal T4	Often none	Treat if TSH > 10, pregnancy, symptoms
Hyperthyroidism (Graves')	**Low TSH**, high T3/T4 (MO 2024); TSH-receptor antibodies	Weight loss with good appetite, heat intolerance, tremor, palpitations/AF, **exophthalmos, pretibial myxoedema** (Graves'-specific), diffuse goitre with bruit	Carbimazole/methimazole; **PTU in the first trimester** (MO 2025); beta-blocker for symptoms; radioiodine; surgery
Toxic multinodular goitre	Low TSH	Elderly, nodules; no eye signs	Radioiodine or surgery

Condition	Labs	Key features	Treatment
Subacute (de Quervain) thyroiditis	Low TSH early, low radioiodine uptake	Painful tender goitre after viral infection, raised ESR	NSAIDs, steroids
Thyroid storm	—	Fever, tachycardia, delirium after surgery/infection	PTU, iodine (after PTU), propranolol, steroids
Congenital hypothyroidism	Neonatal TSH screening	Prolonged jaundice, large tongue, umbilical hernia, delayed milestones — **preventable cause of intellectual disability** (MO 2025)	Early levothyroxine

Thyroid cancers: **papillary commonest** (psammoma bodies, Orphan Annie nuclei, lymphatic spread, radiation); follicular (blood spread); medullary (calcitonin, MEN 2); anaplastic (elderly, worst). Fine-needle aspiration is the first test for a nodule.

Gout

- Cause: hyperuricaemia (uric acid > 7 mg/dL men) from under-excretion (commonest) or over-production; triggers — alcohol (beer), purine-rich meat and seafood, fructose drinks, thiazides, low-dose aspirin, dehydration, surgery.
- Acute attack: sudden, very painful, red hot swollen **first metatarsophalangeal joint (podagra)**, often at night.
- Diagnosis: **needle-shaped, negatively birefringent monosodium urate crystals** in joint fluid (gold standard). Pseudogout = rhomboid, positively birefringent calcium pyrophosphate crystals (knee).
- Chronic: tophi (ear helix, elbows), 'punched-out' erosions with overhanging edges on X-ray, urate kidney stones.
- Treatment: acute — NSAIDs, colchicine, steroids (do not start allopurinol during an attack); long-term — **allopurinol** (xanthine oxidase inhibitor; target uric acid < 6 mg/dL), febuxostat; lifestyle.

Nutritional deficiencies

Deficiency	Disease and signs
Protein–energy	**Kwashiorkor: oedema** (MO 2025), flaky-paint dermatosis, flag sign of hair, fatty liver; **marasmus**: severe wasting, 'old man' face, no oedema; SAM = MUAC < 11.5 cm

Deficiency	Disease and signs
Vitamin A	Night blindness (earliest), Bitot's spots, xerophthalmia, keratomalacia; 1 lakh IU at 9 months, then 2 lakh IU every 6 months to 5 years
Vitamin B1 (thiamine)	Beriberi (wet = heart failure; dry = neuropathy); Wernicke's encephalopathy in alcoholics (give thiamine before glucose)
Vitamin B2	Angular stomatitis, cheilosis, magenta tongue
Niacin (B3)	**Pellagra: dermatitis, diarrhoea, dementia** (death); maize/jowar diets; Casal's necklace
Vitamin B12	Megaloblastic anaemia, **subacute combined degeneration**; strict vegans (MO 2025), pernicious anaemia, metformin
Folate	Megaloblastic anaemia; neural tube defects — 400 µg periconception
Vitamin C	Scurvy: bleeding gums, perifollicular haemorrhage, corkscrew hair, subperiosteal haemorrhage
Vitamin D	Rickets (children: widened wrists, rachitic rosary, bow legs), osteomalacia (adults)
Vitamin K	Bleeding; haemorrhagic disease of newborn — 1 mg IM at birth
Iron	Commonest deficiency: microcytic anaemia, koilonychia, pica; Anemia Mukt Bharat
Iodine	Goitre, cretinism; universal salt iodisation
Zinc	Acrodermatitis enteropathica, poor wound healing, taste loss; ORS + zinc for diarrhoea

Limiting amino acids: cereals (wheat) lack **lysine** (and threonine — MO 2025), pulses lack methionine; mixing them gives complementary protein.

Memory hook: TSH first, always. **P**apillary carcinoma is the **P**opular one (commonest, best outlook, psammoma bodies); **M**edullary **M**akes calcitonin. Gout crystals are **needles** (negatively birefringent) and love the **big toe**.

Eczema, psoriasis and urticaria

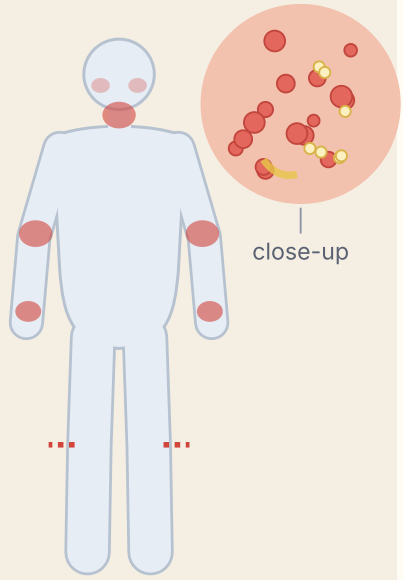
Psoriasis is silvery scales on extensors with Auspitz and Koebner signs; eczema is itchy, flexural and oozing; urticaria is wheals that vanish within 24 hours.

Feature	Eczema (dermatitis)	Psoriasis	Urticaria
Lesion	Itchy papulovesicles, oozing (acute) → lichenified (chronic)	Well-defined erythematous plaques with **silvery-white scales**	Itchy, raised **wheals** with flare; each lasts < 24 h
Sites	Atopic: face/cheeks in infants, **flexures** (elbow, knee folds) in children/adults	**Extensors** (elbows, knees), scalp, lumbosacral area, nails	Anywhere; angio-oedema of lips, eyelids
Signs	Dennie–Morgan fold; white dermographism	**Auspitz sign** (pinpoint bleeding on scraping), **Koebner phenomenon** (lesions at injury sites), candle-grease sign	Dermographism (scratch → wheal)
Nails	Rarely	**Pitting**, onycholysis, oil-drop sign, subungual hyperkeratosis	—
Mechanism	Skin-barrier defect (filaggrin) + type 2 inflammation; contact = type IV	T-cell (Th17, IL-23) driven; rapid epidermal turnover (3–4 days)	Mast-cell histamine release (type I or non-immune)
Associations	Asthma, allergic rhinitis ('atopic march'); raised IgE	**Psoriatic arthritis** (DIP joints, 'pencil-in-cup'), metabolic syndrome	Food, drugs (NSAIDs, penicillin), infections, insect stings; chronic if > 6 weeks
Triggers	Irritants, soaps, wool, heat, sweat	Stress, streptococcal throat (guttate), beta-blockers, lithium, antimalarials, **steroid withdrawal** (pustular)	Food, drugs, cold, pressure, heat, exercise
Treatment	Emollients (mainstay), topical steroids, tacrolimus; antihistamines for itch	Topical steroids, vitamin D analogues (calcipotriol), coal tar, dithranol; phototherapy; methotrexate, biologics	Non-sedating H1 antihistamines (cetirizine); adrenaline for anaphylaxis

Eczema, psoriasis, urticaria

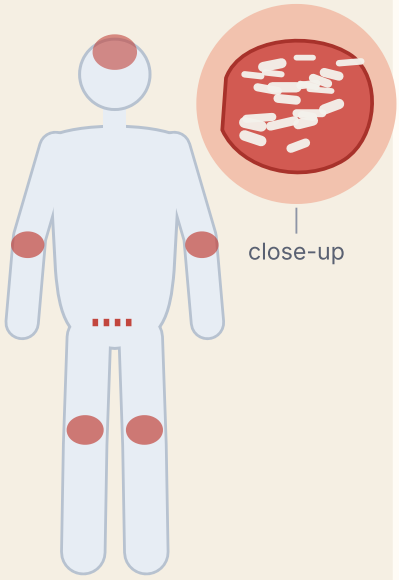
Where it appears, and what the lesion looks like

Eczema (atopic)



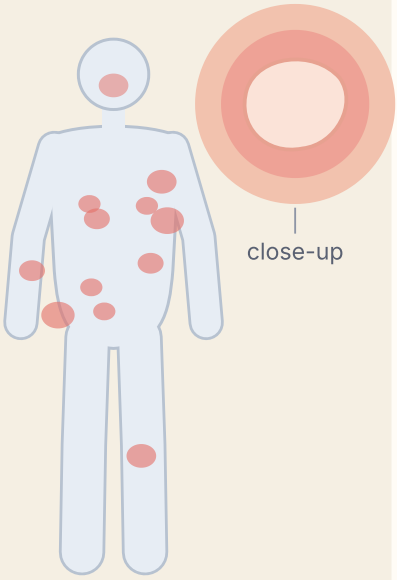
LESION
Itchy papulovesicles, oozing; chronic: thick, lichenified
SITES
FLEXURES: elbow and knee folds; cheeks in infants

Psoriasis



LESION
Well-defined red plaques, silvery-white scales
SITES
EXTENSORS: elbows, knees; scalp, lower back, nails

Urticaria



LESION
Itchy raised wheals with a red flare; each fades < 24 h
SITES
Anywhere; angio-oedema of lips and eyelids

Psoriasis is silvery scales on extensors with Auspitz and Koebner signs; eczema is itchy, flexural and oozing; urticaria is wheals that vanish within 24 hours.

Eczema, psoriasis, urticaria · where each appears and what the lesion looks like

Eczema types: atopic; contact (irritant vs allergic — nickel, the commonest contact allergen; patch test); seborrhoeic (scalp, nasolabial folds, Malassezia; 'cradle cap' in infants); nummular (coin-shaped); stasis (venous, lower legs); dyshidrotic (pompholyx — palms/soles vesicles).

Psoriasis types: plaque (commonest), guttate (drop-like, after strep throat in children), pustular (von Zumbusch — emergency), erythrodermic, inverse (flexural), nail and scalp. Histology: Munro microabscesses, parakeratosis, elongated rete ridges.

Urticaria and anaphylaxis: acute < 6 weeks, chronic > 6 weeks (often idiopathic/autoimmune). **Anaphylaxis** = urticaria with airway or circulatory compromise: **IM adrenaline 0.5 mg (1:1000) into the anterolateral thigh** is the first drug; then fluids, antihistamine, steroid.

- Other skin MCQs: **scabies** — burrows in finger webs, wrists, genitals; night itch; family members affected; **permethrin 5%**.
- **Tinea** — annular, scaly, active edge; KOH mount.
- **Vitiligo** — depigmented patches, Wood's lamp; Koebner positive.
- **Lichen planus** — '6 Ps' (purple, polygonal, pruritic, planar papules), Wickham's striae.
- **Pemphigus vulgaris** — flaccid intraepidermal blisters, Nikolsky positive; **bullous pemphigoid** — tense subepidermal blisters, elderly.
- **Leprosy** — hypopigmented anaesthetic patches, thickened nerves (ulnar, common peroneal).

Memory hook: pemphig**us** blisters are flaccid and burst on **us** (Nikolsky sign positive, high in the skin); pemphig**oid** blisters are tense and sit deep. Lichen planus is the **6 Ps**: purple, pruritic, polygonal, planar papules and plaques.

Exanthematous fevers and seasonal outbreaks

The order in which rashes appear (chickenpox day 1, scarlet fever day 2, smallpox day 3, measles day 4, typhus day 5) and the monsoon fevers of Goa are the most-asked items.

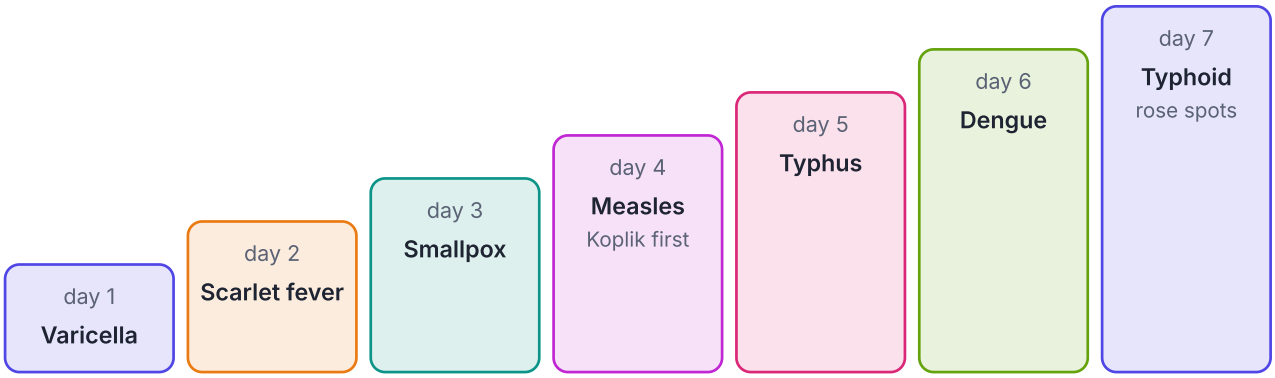
Exanthematous fevers

Disease	Agent	Rash and clues	Complication / note
Measles	Paramyxovirus	Prodrome of the 3 Cs (cough, coryza, conjunctivitis); **Koplik spots** on buccal mucosa; maculopapular rash on day 4, behind ears → face → trunk, leaves brown staining	Pneumonia (commonest cause of death), diarrhoea, SSPE years later; vitamin A; MR vaccine at 9–12 and 16–24 months
Rubella (German measles)	Togavirus	Mild; rash day 1; **postauricular/suboccipital lymph nodes**	Congenital rubella: **PDA**, cataract, deafness (MO 2024)
Chickenpox	Varicella-zoster	**Pleomorphic rash** (macule, papule, vesicle, crust at once), centripetal (trunk), 'dewdrop on a rose petal', spares palms/soles	Secondary infection, pneumonia (adults); zoster later; acyclovir
Smallpox (eradicated 1980)	Variola	Centrifugal, all lesions at the same stage, palms and soles involved	Last case India 1975

Disease	Agent	Rash and clues	Complication / note
Scarlet fever	Group A Streptococcus	Sandpaper rash, **strawberry tongue**, circumoral pallor, Pastia's lines, desquamation	Rheumatic fever, glomerulonephritis
Roseola infantum	HHV-6	High fever 3–4 days, rash appears **as fever falls**	Febrile seizures
Erythema infectiosum (fifth disease)	Parvovirus B19	'**Slapped cheek**' then lacy rash	Aplastic crisis in sickle cell; hydrops fetalis
Hand, foot and mouth disease	Coxsackie A16, enterovirus 71	Oral ulcers + vesicles on palms and soles	Common in Indian toddlers, monsoon
Dengue	Flavivirus	Fever, headache, retro-orbital pain, myalgia ('breakbone'), flushing; convalescent rash 'isles of white in a sea of red'	See below

The rash staircase: which fever shows its rash on which day

"Very Sick Persons Must Take Double Tablets", one step per day of fever



rash staircase · day 1 to day 7

Climb the stairs one day at a time: the later the rash, the higher the step.

Seasonal outbreaks in Goa (monsoon and post-monsoon)

Disease	Spread	Clinical clue	Key test	Drug / control
Dengue	**Aedes aegypti** (day-biter, clean stored water)	Fever, myalgia, rash, low platelets; warning signs (abdominal pain, vomiting, bleeding, lethargy, rising haematocrit)	**NS1 (day 1–5)**, IgM ELISA (after day 5)	Fluids, paracetamol; **avoid NSAIDs/aspirin**; source reduction
Chikungunya	Aedes	Fever with severe, lasting polyarthralgia	IgM ELISA / RT-PCR	Symptomatic
Malaria	Female Anopheles (An. stephensi in urban Goa)	Periodic fever with chills	Smear, RDT	Vivax: chloroquine + primaquine 14 days; falciparum: ACT (artesunate + sulfadoxine–pyrimethamine; artemether–lumefantrine in NE states); severe: IV artesunate
Leptospirosis	Rat urine in water/soil (paddy fields, floods)	Conjunctival suffusion, calf pain, jaundice + renal failure (Weil's)	IgM ELISA, MAT	Doxycycline/penicillin; doxycycline prophylaxis 200 mg weekly
Scrub typhus	Larval mite (chigger)	**Eschar**, fever, lymphadenopathy	IgM ELISA	Doxycycline
Typhoid	Faeco-oral	Step-ladder fever, relative bradycardia, coated tongue	Blood culture (week 1)	Ceftriaxone, azithromycin
Cholera / acute gastroenteritis	Water	Rice-water stools	Stool culture	**ORS** + zinc (children); doxycycline
Hepatitis A and E	Water	Jaundice outbreaks	IgM antibodies	Supportive; safe water
Influenza (H1N1)	Droplets	Fever, cough, sore throat	RT-PCR	Oseltamivir for high-risk (category B/C)
Kyasanur Forest Disease	Tick (Haemaphysalis); monkey deaths	Fever, bleeding, later neurological phase; Sattari taluka	RT-PCR	KFD vaccine in endemic areas

Disease	Spread	Clinical clue	Key test	Drug / control
Japanese encephalitis	Culex; pigs amplify	Encephalitis in children	IgM ELISA (CSF)	JE vaccine

Dengue classification (WHO 2009): dengue without warning signs, dengue with warning signs, severe dengue (shock from plasma leakage, severe bleeding, organ impairment). Rising haematocrit ≥ 20% with falling platelets = plasma leakage. Platelet transfusion only for bleeding or very low counts.

Genus epidemicus: in an outbreak, the homoeopath studies many cases to find the remedy covering most of them (§100–102), then uses it for treatment and prophylaxis — for example Eupatorium perfoliatum in dengue-like fevers, Arsenicum album was promoted by AYUSH as prophylaxis during COVID-19.

Memory hook: the day the rash appears: "**V**ery **S**ick **P**ersons **M**ust **T**ake **D**ouble **T**ablets": **V**aricella day 1, **S**carlet fever day 2, small**P**ox day 3, **M**easles day 4, **T**yphus day 5, **D**engue day 6, **T**yphoid day 7. Koplik spots come before the measles rash.

Homoeopathic therapeutics: leading remedies by disease

GPSC asks a disease plus a keynote and expects one remedy. Learn the keynote column; the remedy follows from it. Prescribe always on the totality — these are exam pointers, not a substitute for individualisation.

Disease	Remedy	Keynote that identifies it
Hypertension	Glonoinum	Bursting, throbbing head; worse sun and heat; surging of blood
Hypertension	Aurum metallicum	High BP with depression, suicidal thoughts, arteriosclerosis
Hypertension (mother tincture)	Rauwolfia serpentina	Organ remedy for high BP
Angina pectoris	Cactus grandiflorus	Heart constricted as if by an iron band
Angina pectoris	Latrodectus mactans	Violent precordial pain to left arm and fingers, numbness, anxiety

Disease	Remedy	Keynote that identifies it
Heart failure / weak heart	Digitalis	Very slow, irregular, weak pulse; fears heart will stop if he moves
Heart tonic (mother tincture)	Crataegus oxyacantha	Weak myocardium, irregular pulse
Bronchial asthma	Arsenicum album	Worse after midnight (1–3 a.m.), restless, anxious, must sit up, better warmth
Bronchial asthma	Ipecacuanha	Wheeze with constant nausea, rattling but nothing comes up
Asthma (hydrogenoid)	Natrum sulphuricum	Worse damp weather and at 4–5 a.m.; loose morning stool
COPD / chronic bronchitis	Antimonium tartaricum	Coarse rattling, little expectoration, drowsy, cyanosed (elderly)
COPD / air hunger	Carbo vegetabilis	Wants to be fanned, cold breath, collapse
Stroke / paralysis	Causticum	Gradual paralysis, right-sided, hoarseness, sympathetic
Stroke (haemorrhage)	Arnica montana	Bruised feeling, apoplexy, bed feels too hard
Stroke (apoplectic stupor)	Opium	Stertorous breathing, red face, painless
Epilepsy	Cuprum metallicum	Violent cramps, aura from knees; begins in fingers and toes
Epilepsy	Cicuta virosa	Opisthotonos, after head injury
Epilepsy	Bufo rana	Fits during sleep or with sexual excitement; foolish
Peptic ulcer / acidity	Nux vomica	Sedentary, stimulants, irritable; ineffectual urging
Peptic ulcer	Argentum nitricum	Craves sweets which disagree; belching; ulcer pain radiates
Peptic ulcer (bleeding)	Kali bichromicum	Round punched-out ulcers; pain in small spots
GERD / heartburn	Robinia pseudacacia	Intense sour acidity, worse at night
IBS	Argentum nitricum	Diarrhoea from anticipation, sweets

Disease	Remedy	Keynote that identifies it
IBS	Lycopodium	Bloating 4–8 p.m., flatulence, anxiety about performance
IBS / diarrhoea	Aloe socotrina	Morning urging, insecurity of rectum
Hepatitis / jaundice	Chelidonium majus	Pain under the right scapula; liver remedy
Hepatitis	Phosphorus	Craving for cold water, bleeding, fatty liver
Cirrhosis	Lycopodium	Liver remedy with ascites, right-sided
Gallstone colic	Berberis vulgaris / Calcarea carb	Radiating pains (Berberis); stone diathesis
Cholecystitis	Bryonia alba	Worse slightest motion, better pressure, thirst
Diabetes (mother tincture)	Syzygium jambolanum	Organ remedy; lowers glucose
Diabetes	Uranium nitricum	Great emaciation, debility, polyuria
Diabetes	Phosphoric acid	Diabetes after grief, milky urine at night, apathy
Hypothyroidism	Calcarea carbonica	Fat, flabby, chilly, sweaty head, slow
Hyperthyroidism	Iodium	Ravenous hunger yet emaciation, hot, restless
Hyperthyroidism / exophthalmos	Lycopus virginicus	Palpitation with exophthalmic goitre
Gout	Colchicum autumnale	Great toe, nausea from the smell of food, worse touch
Gout	Ledum palustre	Ascending gout, better cold applications
Gout (chronic, uric acid)	Urtica urens	Uric acid diathesis, burning itch
Atopic eczema	Graphites	Honey-like sticky discharge, cracks in folds
Eczema / itch	Sulphur	Burning itch worse warmth and washing
Eczema with crusts	Mezereum	Thick crusts with pus beneath
Psoriasis	Arsenicum album	Dry, scaly, burning, better warmth
Psoriasis	Petroleum	Deep cracks, worse winter

Disease	Remedy	Keynote that identifies it
Urticaria	Apis mellifica	Stinging, burning wheals, better cold
Urticaria	Urtica urens	Burning, stinging wheals after shellfish
Urticaria	Rhus toxicodendron	Urticaria from getting wet, restless
Measles	Pulsatilla	Bland catarrh, thirstless, weepy; also Euphrasia for eye signs
Measles (early)	Aconite / Belladonna	Sudden high fever
Chickenpox	Rhus toxicodendron / Antimonium crudum	Itchy vesicles; restless (Rhus)
Mumps	Pilocarpinus / Mercurius	Parotitis with salivation
Dengue / bone-breaking fever	Eupatorium perfoliatum	Bone-breaking pains
Influenza	Gelsemium	Drowsy, dull, trembling, thirstless
Influenza / cold	Arsenicum album	Watery burning coryza, restless, anxious
Malaria	China officinalis	Periodicity, weakness from loss of fluids
Malaria	Natrum muriaticum	Chill at 10 a.m., bursting headache
Typhoid	Baptisia tinctoria	Besotted face, body feels scattered
Leptospirosis / jaundice with myalgia	Bryonia, Phosphorus	By totality

The chief epidemic-prophylaxis examples GPSC can ask: **Belladonna** for scarlet fever (Hahnemann, 1801), **Eupatorium** for dengue-like fevers, **Arsenicum album 30** promoted by the AYUSH ministry for COVID-19, and **Influenzinum** for flu.

Memory hook: the **genus epidemicus** pairs to learn: **B**elladonna for scarlet fever ("**B**right red"), **E**upatorium for dengue ("break-bon**E**"), **A**rsenicum album for the AYUSH COVID-19 advisory, **I**nfluenzinum for flu.

Rapid revision: 60 one-liners

Read this the night before and on the morning of the exam.

Numbers

1. Hypertension ≥ 140/90; target in diabetes < 130/80.
2. Hypertensive emergency ≥ 180/120 with organ damage; lower by ≤ 25% in the first hour.
3. Salt < 5 g/day (WHO).
4. Diabetes: FPG ≥ 126, 2-h OGTT ≥ 200, HbA1c ≥ 6.5%, random ≥ 200 with symptoms.
5. Prediabetes: FPG 100–125, 2-h 140–199, HbA1c 5.7–6.4%.
6. HbA1c target < 7%; it reflects 2–3 months.
7. Hypoglycaemia < 70 mg/dL.
8. COPD: post-bronchodilator FEV1/FVC < 0.70.
9. Asthma reversibility: FEV1 ↑ ≥ 12% and ≥ 200 mL.
10. COPD oxygen target 88–92%; LTOT if PaO2 ≤ 55 mmHg, ≥ 15 h/day.
11. Stroke thrombolysis within 4.5 h; status epilepticus > 5 min.
12. GCS 3–15; ≤ 8 = intubate.
13. Uric acid target < 6 mg/dL on allopurinol.
14. SAM: MUAC < 11.5 cm.
15. Microalbuminuria 30–300 mg/day.

Drugs of choice

16. Acute asthma: salbutamol (inhaled SABA).
17. Asthma controller: inhaled corticosteroid.
18. Anaphylaxis: IM adrenaline 0.5 mg (1:1000).
19. Type 2 diabetes first line: metformin; pregnancy: insulin.
20. DKA: IV normal saline first, then insulin; watch potassium.
21. Hyperthyroidism in first trimester: propylthiouracil.
22. Hypothyroidism: levothyroxine on an empty stomach.
23. Absence seizures: ethosuximide; status epilepticus: IV lorazepam.
24. Hepatic encephalopathy: lactulose (+ rifaximin).
25. Ascites: spironolactone.
26. Variceal bleed: octreotide/terlipressin + band ligation; prevention: propranolol.
27. Acute gout: NSAIDs/colchicine; long-term: allopurinol.
28. H. pylori: PPI + clarithromycin + amoxicillin.
29. Scabies: permethrin 5%.
30. Scrub typhus and leptospirosis: doxycycline.

31. Paracetamol poisoning: N-acetylcysteine.
32. Hypertension in pregnancy: labetalol, nifedipine, methyldopa (no ACE inhibitors).

Investigations of choice

33. Acute stroke / SAH: non-contrast CT.
34. Gallstones: ultrasound.
35. GERD gold standard: 24-h pH study.
36. Bronchiectasis: HRCT.
37. Gout: urate crystals in joint fluid.
38. Thyroid screening: TSH; thyroid nodule: FNAC.
39. MI: troponin.
40. Pulmonary TB (NTEP): NAAT (CBNAAT/TrueNat).
41. Dengue day 1–5: NS1.
42. Heart failure marker: BNP.

Signs and associations

43. Koplik spots = measles; strawberry tongue = scarlet fever; slapped cheek = parvovirus B19.
44. Rash on day 1 chickenpox, day 4 measles; rash as fever falls = roseola.
45. Auspitz sign, Koebner phenomenon, nail pitting = psoriasis.
46. Murphy's sign = acute cholecystitis; Charcot's triad = cholangitis.
47. Asterixis = hepatic encephalopathy (also CO_2 retention, uraemia).
48. Kussmaul breathing + fruity breath = DKA.
49. Exophthalmos + pretibial myxoedema = Graves'.
50. Podagra = gout; needle-shaped negatively birefringent crystals.
51. Pellagra = dermatitis, diarrhoea, dementia.
52. Dry cough with ACE inhibitors; ankle oedema with amlodipine.

⚠ Traps

53. Type 1, not type 2, diabetes is ketosis-prone.
54. Duodenal ulcer pain is relieved by food; gastric ulcer pain worsens.
55. Bell's palsy involves the forehead; stroke (UMN) spares it.
56. Valproate is the most teratogenic antiepileptic; carbamazepine worsens myoclonic epilepsy.
57. Avoid aspirin/NSAIDs in dengue.

58. Do not start allopurinol during an acute gout attack.

59. Rhus tox: better from continued motion; Bryonia: worse from any motion.

60. Arsenicum asthma is after midnight; Natrum sulph asthma is 4–5 a.m. in damp weather.

One-glance recall card

Topic 7 recall card: the 15 facts asked most

Cover the small lines, say the fact from the big cue, then check

140/90 hypertension on repeated readings	**ACE-i cough** dry cough from bradykinin; switch to an ARB	**Troponin** rises in hours, stays about 1 to 2 weeks
Reversible asthma comes and goes; COPD persists	**FEV1/FVC < 0.7** spirometry shows airflow obstruction	**Plain CT first** rules out bleeding before thrombolysis
Forehead spared upper motor neurone; Bell's palsy takes it all	**Food helps?** duodenal ulcer eases with food; gastric worsens	**AST:ALT > 2** alcoholic liver disease; viral: ALT higher
126 · 200 · 6.5 fasting, 2-hour, HbA1c cut-offs	**TSH first** best single thyroid test; papillary commonest	**Needle crystals** gout: negatively birefringent, big toe
6 Ps lichen planus: purple, pruritic, polygonal...	**VSPMTDT** rash days 1-7: varicella, scarlet ... typhoid	**Genus epidemicus** Bell scarlet, Eup dengue, Ars COVID, Influenzinum

Topic 7 recall card · 15 tiles

For the last ten minutes before the exam: cover the small lines of each tile and say the fact from the big cue.

Practice MCQs (50)

Time yourself: 50 questions in 35 minutes. Cover the key until you finish.

1. Hypertension is diagnosed at a repeated clinic blood pressure of at least: (A) 120/80 mmHg (B) 130/85 mmHg (C) 140/90 mmHg (D) 160/100 mmHg
2. The BP goal for a diabetic with hypertension is: (A) below 130/80 mmHg (B) below 150/90 mmHg (C) below 160/100 mmHg (D) below 110/70 mmHg

3. A dry, persistent cough is a typical side-effect of: (A) amlodipine (B) ACE inhibitors (C) thiazides (D) beta-blockers
4. Which antihypertensive is contraindicated in pregnancy? (A) labetalol (B) nifedipine (C) methyldopa (D) enalapril
5. The most specific biomarker of myocardial infarction is: (A) CK total (B) LDH (C) troponin (D) AST
6. ST elevation in leads II, III and aVF indicates infarction of the: (A) anterior wall (B) lateral wall (C) septum only (D) inferior wall
7. The commonest cause of death in the first hours after MI is: (A) cardiogenic shock (B) free-wall rupture (C) Dressler syndrome (D) ventricular fibrillation
8. Furosemide is indicated in: (A) hypokalaemia (B) gout (C) acute pulmonary oedema (D) hypovolaemic shock
9. COPD is confirmed by a post-bronchodilator FEV1/FVC ratio below: (A) 0.70 (B) 0.90 (C) 0.85 (D) 0.50
10. The drug for an acute asthma attack is inhaled: (A) budesonide alone (B) montelukast (C) theophylline (D) salbutamol
11. Panacinar emphysema in a young non-smoker suggests: (A) cystic fibrosis (B) asthma (C) alpha-1 antitrypsin deficiency (D) silicosis
12. Interventions proven to improve survival in COPD are smoking cessation and: (A) inhaled steroids (B) long-term oxygen therapy (C) antibiotics (D) theophylline
13. In a COPD exacerbation, the oxygen saturation target is: (A) 88–92% (B) 98–100% (C) 80–84% (D) 94–98%
14. The commonest risk factor for stroke is: (A) diabetes (B) hypertension (C) smoking (D) alcohol
15. The first investigation in suspected acute stroke is: (A) MRI angiography (B) EEG (C) lumbar puncture (D) non-contrast CT brain
16. IV thrombolysis for ischaemic stroke is given within: (A) 12 hours (B) 24 hours (C) 4.5 hours (D) 48 hours
17. Thunderclap headache with neck stiffness most suggests: (A) migraine (B) tension headache (C) sinusitis (D) subarachnoid haemorrhage
18. A cardiac source of embolic stroke in rheumatic heart disease is usually: (A) atrial fibrillation (B) ventricular septal defect (C) aortic stenosis (D) pericarditis
19. The drug of choice for absence seizures is: (A) phenytoin (B) carbamazepine (C) phenobarbitone (D) ethosuximide
20. The first drug for status epilepticus is IV: (A) phenytoin (B) lorazepam (C) valproate (D) levetiracetam
21. The commonest cause of acquired epilepsy in India is: (A) tuberculoma (B) neurocysticercosis (C) head injury (D) brain tumour
22. Bell's palsy differs from stroke facial weakness because it: (A) involves the forehead (B) spares the forehead (C) is always bilateral (D) has no taste loss

23. Epigastric pain relieved by food and waking the patient at night suggests: (A) duodenal ulcer (B) gastric ulcer (C) GERD (D) cholecystitis
24. The best non-invasive test to confirm H. pylori eradication is: (A) serology (B) barium meal (C) urea breath test (D) CT
25. The main mechanism of GERD is: (A) LES hypertension (B) transient lower oesophageal sphincter relaxation (C) hiatus hernia alone (D) delayed gastric emptying
26. The gold standard test for GERD is: (A) 24-hour pH monitoring (B) barium swallow (C) endoscopy (D) manometry
27. A red-flag feature that argues against IBS is: (A) rectal bleeding (B) pain relieved by defecation (C) bloating (D) stress-related symptoms
28. Hepatitis with the highest mortality in pregnancy is: (A) hepatitis A (B) hepatitis B (C) hepatitis C (D) hepatitis E
29. Asterixis in a cirrhotic patient indicates: (A) hepatic encephalopathy (B) portal vein thrombosis (C) SBP (D) hepatorenal syndrome
30. The drug of first choice for cirrhotic ascites is: (A) furosemide alone (B) mannitol (C) spironolactone (D) acetazolamide
31. Murphy's sign is positive in: (A) acute cholecystitis (B) acute appendicitis (C) pancreatitis (D) peptic ulcer
32. The investigation of choice for gallstones is: (A) plain X-ray (B) ultrasonography (C) CT (D) HIDA scan
33. Diabetes is diagnosed by an HbA1c of at least: (A) 5.7% (B) 7.5% (C) 6.5% (D) 8.0%
34. The diabetic emergency most typical of type 1 diabetes is: (A) hyperosmolar state (B) diabetic ketoacidosis (C) lactic acidosis (D) hypothyroid coma
35. The first step in treating DKA is: (A) IV insulin bolus alone (B) IV normal saline (C) IV bicarbonate (D) oral glucose
36. The first-line oral drug for type 2 diabetes is: (A) glimepiride (B) pioglitazone (C) acarbose (D) metformin
37. The earliest sign of diabetic nephropathy is: (A) raised creatinine (B) haematuria (C) microalbuminuria (D) oedema
38. In primary hypothyroidism the thyroid profile shows: (A) low TSH, high T4 (B) low TSH, low T4 (C) high TSH, low T4 (D) normal TSH, high T4
39. The antithyroid drug preferred in the first trimester is: (A) carbimazole (B) methimazole (C) propylthiouracil (D) radioactive iodine
40. Exophthalmos and pretibial myxoedema are specific to: (A) toxic multinodular goitre (B) thyroid adenoma (C) Hashimoto's thyroiditis (D) Graves' disease
41. Needle-shaped negatively birefringent crystals in joint fluid are diagnostic of: (A) pseudogout (B) rheumatoid arthritis (C) gout (D) septic arthritis
42. Allopurinol acts by inhibiting: (A) uricase (B) xanthine oxidase (C) cyclo-oxygenase (D) tubular secretion

13. Dermatitis, diarrhoea and dementia indicate deficiency of: (A) thiamine (B) riboflavin (C) vitamin B12 (D) niacin
14. Silvery scales with pinpoint bleeding on scraping (Auspitz sign) indicate: (A) psoriasis (B) eczema (C) lichen planus (D) tinea
15. The first drug in anaphylaxis is: (A) IV hydrocortisone (B) IM adrenaline (C) oral cetirizine (D) IV fluids alone
16. Koplik spots are seen in: (A) measles (B) rubella (C) chickenpox (D) scarlet fever
17. A pleomorphic vesicular rash, centripetal and sparing palms and soles, is: (A) smallpox (B) chickenpox (C) measles (D) hand, foot and mouth disease
18. Drug to be avoided in dengue fever: (A) paracetamol (B) aspirin (C) ORS (D) IV fluids
19. For dengue with bone-breaking pains, the leading homoeopathic remedy is: (A) Bryonia (B) Rhus tox (C) Arnica (D) Eupatorium perfoliatum
20. Asthma worse after midnight with anxiety, restlessness and relief from warmth calls for: (A) Arsenicum album (B) Natrum sulph (C) Ipecac (D) Antimonium tart

Answer key

Q	Ans	Why
1	C	Indian and ISH guidelines
2	A	GPSC MO 2024 keyed 120–130/70–79
3	B	Bradykinin accumulation
4	D	ACE inhibitors and ARBs are teratogenic
5	C	Rises in 3–4 h, stays 7–10 days
6	D	Usually the right coronary artery
7	D	Arrhythmia
8	C	Loop diuretic (GPSC MO 2024)
9	A	GOLD criterion
10	D	Short-acting beta-2 agonist
11	C	Lower lobes
12	B	LTOT ≥ 15 h/day

Q	Ans	Why
13	A	Avoid CO2 retention
14	B	Most important modifiable risk
15	D	Excludes haemorrhage before thrombolysis
16	C	Alteplase or tenecteplase
17	D	Ruptured berry aneurysm
18	A	GPSC MO 2025 case
19	D	Or valproate; 3 Hz spike-wave EEG
20	B	Benzodiazepine first
21	B	Taenia solium
22	A	Lower motor neurone lesion
23	A	Hunger pain
24	C	Or stool antigen
25	B	GPSC MO 2024
26	A	Endoscopy for alarm features
27	A	Also weight loss, nocturnal diarrhoea, age > 50
28	D	Fulminant failure, about 20%
29	A	Flapping tremor
30	C	Aldosterone antagonist
31	A	Inspiration arrested on RUQ palpation
32	B	Most stones are not radio-opaque
33	C	5.7–6.4% = prediabetes
34	B	GPSC MO 2025 trap
35	B	Then insulin infusion and potassium
36	D	No hypoglycaemia

Q	Ans	Why
37	C	30–300 mg/day
38	C	Hashimoto's commonest where iodine is sufficient
39	C	GPSC MO 2025
40	D	TSH-receptor antibodies
41	C	Monosodium urate
42	B	Start after the acute attack settles
43	D	Pellagra
44	A	Extensor plaques
45	B	0.5 mg of 1:1000 in the thigh
46	A	Buccal mucosa, before the rash
47	B	Lesions at different stages
48	B	Bleeding risk
49	D	Bone-breaking pains
50	A	1–3 a.m. aggravation

Score guide: 40+ is exam-ready; below 30, reread the tables for the questions you missed.

Sources

GPSC question papers with answer keys (questions quoted in shortened form):

- Medical Officer CBRT, 28 Sep 2025
- Medical Officer CBRT, 28 Sep 2024
- GPSC question paper archive

Standard texts the notes follow (facts written from them, not copied; check a doubtful figure against your own edition):

- *Davidson's Principles and Practice of Medicine* — cardiology, respiratory, GI, endocrine, infections.
- *API Textbook of Medicine* — Indian context: tropical fevers, neurocysticercosis, snake bite.
- *Harrison's Principles of Internal Medicine* — definitions and cut-offs.

- Guidelines: ISH 2020 hypertension, GINA and GOLD (asthma, COPD), ADA standards of care (diabetes), WHO 2009 dengue classification, Indian national programme treatment guidelines (malaria, TB).
- Homoeopathic therapeutics: Boericke's *Pocket Manual of Homoeopathic Materia Medica with Repertory*, Kent's *Lectures on Materia Medica*, Clarke's *Dictionary of Practical Materia Medica*.

Guideline numbers are as of mid-2026; if a question words a cut-off differently, follow the Indian guideline it names.